hovers at a fixed height above the horizon. At the transition from blue to green, time flows 20 percent as fast as it flows far from the hole. At the transition from yellow to red, time is slowed to 10 percent of its normal rate far away. And at the black circle, the bottom of the surface, time slows to a halt. This is the event horizon. It is a circle, not a sphere, because we are looking only at the equatorial plane, only at two dimensions of our universe (of our brane). If we were to restore the third space dimension, the horizon would become a flattened sphere: a spheroid. The white arrows depict the rate at which space whirls around the black hole. The whirl is fast at the horizon, and decreases as we climb upward in a spacecraft.

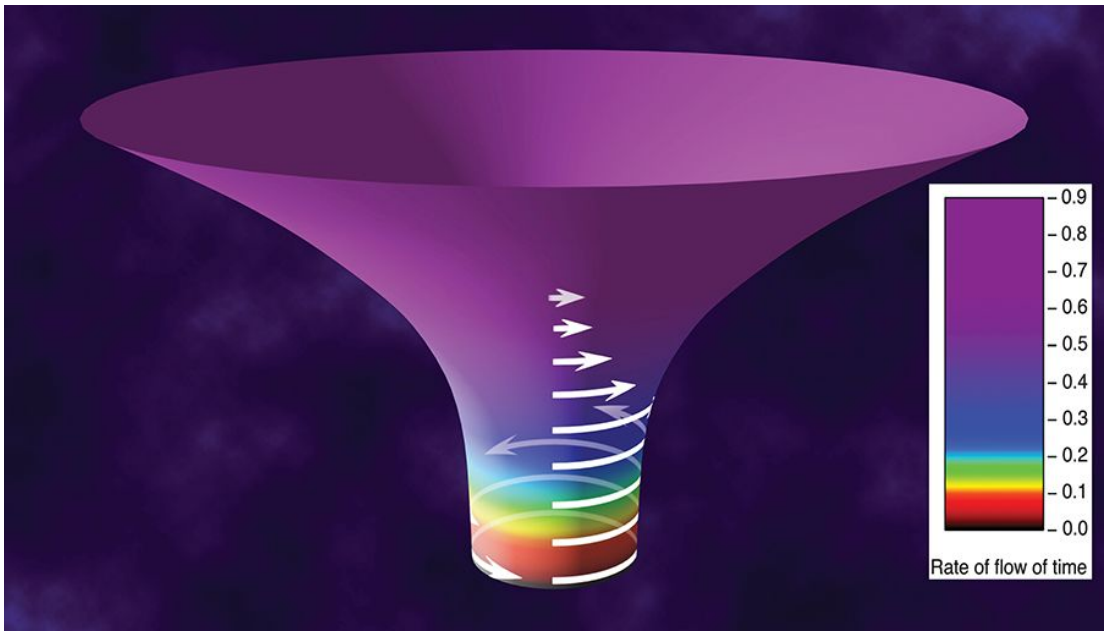


Fig. 5.5. Precise depiction of the warped space and time around a rapidly spinning black hole: one that spins at 99.8 percent of the maximum possible rate. *[Drawing by Don Davis based on a sketch by me.]*

In the fully accurate Figure 5.5, I don't depict the hole's interior. We'll get to that later, in Chapters 26 and 28.

The warping in Figure 5.5 is the essence of a black hole. From its details, expressed mathematically, physicists can deduce everything